

# **Automated Epileptic Seizure Detection Using EEG Signals and XGBoost**

*Project Documentation*

## **1. Abstract**

Epileptic seizures are sudden bursts of abnormal electrical activity in the brain and early detection is important for timely intervention. Manual analysis of EEG recordings requires specialist knowledge and can be time-consuming. This project develops an automated seizure detection approach using EEG signals from the Bonn EEG dataset. The system extracts time-domain statistics, Hjorth parameters, entropy, wavelet energy, and frequency-domain information from EEG signals. These features are used as inputs to an XGBoost classifier for binary seizure and non-seizure classification. The application supports both file-based EEG input and manual data entry and displays the prediction together with an EEG signal graph.

## **2. Introduction**

Epilepsy is a neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain. EEG signals provide a non-invasive way to record this activity. The project focuses on automating part of the EEG analysis process by transforming raw signals into meaningful numerical features and applying machine learning for classification. The system is designed as an interactive application that supports both uploaded EEG files and manual signal entry.

## **3. Problem Statement**

Manual EEG analysis for seizure detection is time-consuming and depends on specialist knowledge. There is a need for an automated approach that can analyze EEG signals efficiently, identify useful signal characteristics, and classify the input as seizure or non-seizure. The system should also provide a convenient interface for file-based and manual input.

## **4. Existing System**

Conventional seizure detection mainly depends on neurologists examining EEG recordings. Continuous monitoring systems may assist with monitoring but can produce false alarms when signals contain noise or disturbances. These limitations motivate the development of an automated machine-learning-based approach.

## 5. Proposed System

The proposed system combines EEG signal processing, feature extraction, and XGBoost classification. Users log in, provide EEG data through a file or manual entry, and the system extracts time-domain, Hjorth, entropy, wavelet, and FFT-related features. The trained XGBoost model predicts seizure or non-seizure activity, while the application displays the EEG graph and result.

## 6. Dataset

The project uses the Bonn EEG dataset as described in the project presentation. It contains five groups A, B, C, D, and E. Groups A and B represent healthy subjects, groups C and D represent seizure-free patients, and group E contains epileptic seizure activity. The project description specifies 500 EEG segments in total, 4097 samples per segment, and approximately 23.6 seconds per segment. Nineteen extracted features are used for machine-learning classification.

## 7. Feature Extraction

The system extracts mean, standard deviation, maximum, minimum, signal energy, Hjorth activity, Hjorth mobility, Hjorth complexity, entropy, wavelet energy using the db4 wavelet, and FFT-based frequency features. These features transform raw EEG signals into numerical representations suitable for machine-learning classification.

## 8. XGBoost Algorithm

XGBoost is used as the classification algorithm. It builds decision trees sequentially. Each new tree focuses on reducing errors from the preceding model, and the contributions of the trees are combined to produce the final prediction. For this project, the classifier is trained to distinguish seizure and non-seizure EEG signals.

## 9. System Workflow

The workflow consists of user authentication, EEG data input, feature extraction, feature normalization where applicable, XGBoost prediction, and result display. The result includes the predicted seizure status and a graph of the uploaded or entered EEG signal.

## 10. Functional Requirements

The system provides registration and login, EEG file upload, manual EEG input, feature extraction, XGBoost classification, prediction display, EEG signal visualization, input-error handling, and storage of user details in SQLite.

## 11. Non-Functional Requirements

The system should provide reliable predictions, reasonable response time, simple usability, maintainability, security of user information, and compatibility with supported Python environments and modern browsers.

## 12. Technology Stack

Python 3.x, Streamlit, NumPy, Pandas, SciPy, Scikit-learn, XGBoost, PyWavelets, Matplotlib, Joblib, and SQLite are used in the project.

## 13. Testing

Testing includes valid seizure and non-seizure EEG inputs, file upload, manual input, invalid input handling, user registration and login, EEG graph generation, and prediction result display.

## 14. Conclusion

The project demonstrates an automated approach for EEG seizure classification by combining signal processing, feature extraction, and XGBoost. Support for file upload and manual input makes the system useful for academic testing and demonstration, while signal visualization helps users inspect the input alongside the prediction. The system is intended for research and educational use and should not replace professional medical diagnosis.

## 15. References

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